

Topic : CROP YIELD PREDICTION AND CLIMATE RISK ASSESSMENT SYSTEM
“Predict. Prepare. Prosper”

By Department of Data Science



Presented By:

- | | | |
|------------|---|------------------|
| 23VE1A6772 | - | D. VAISHNAVI |
| 23VE1A6781 | - | S. JAYESH |
| 23VE1A6799 | - | Shaik Aboobakr |
| 23VE1A67A4 | - | U. Chelsea Reddy |

Abstract

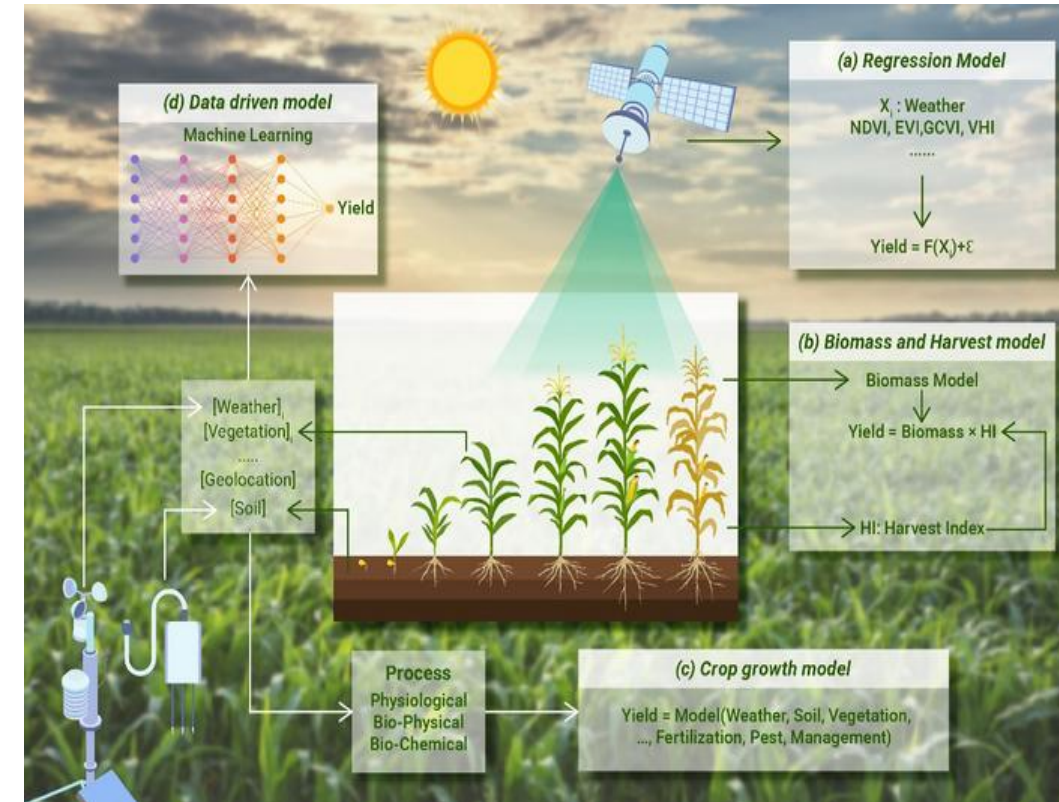
- Agriculture is highly dependent on climatic conditions, and increasing climate variability has made crop yield prediction more challenging and uncertain. Irregular rainfall, rising temperatures, droughts, and extreme weather events significantly impact agricultural productivity, especially in climate-sensitive regions.
- This project presents a Machine Learning-based system for crop yield prediction integrated with climate risk assessment to support data-driven agricultural decision-making.
- The proposed system utilizes historical crop yield data along with key climatic parameters such as rainfall, temperature, humidity, soil moisture, and vegetation indices (NDVI). Various machine learning algorithms including Random Forest, XGBoost, and LSTM are employed to model both linear and non-linear relationships between environmental factors and crop performance.
- In addition to predicting expected yield, the system calculates a climate risk score by analyzing deviations in rainfall, temperature extremes, and drought indicators. The final output provides predicted yield values, categorized risk levels, and interpretable insights into the most influential climatic factors.
- This integrated approach enables proactive planning, optimized resource allocation, and improved agricultural resilience. The system aims to reduce uncertainty in crop production and contribute to sustainable and climate-aware farming practices.

Introduction

- Agriculture plays a vital role in the economy and food security of countries like India, where a large portion of the population depends directly or indirectly on farming for livelihood. However, agricultural productivity is highly sensitive to changing climatic conditions.
- In recent years, climate variability characterized by irregular rainfall patterns, increasing temperatures, prolonged droughts, floods, and other extreme weather events has created significant uncertainty in crop production. These unpredictable environmental changes make traditional methods of yield estimation less reliable and increase risks for farmers and policymakers.
- Accurate crop yield prediction is essential for effective agricultural planning, market stabilization, food supply management, and resource optimization. Conventional forecasting approaches often rely on historical averages and manual observations, which may not adequately capture complex and non-linear relationships between climatic factors and crop growth.
- With the advancement of Artificial Intelligence (AI) and Machine Learning (ML), data-driven techniques now offer more precise and adaptive solutions for agricultural forecasting. Crop Yield Prediction and Climate Risk Assessment System aims to address these challenges by integrating machine learning models with climate analytics.
- The system leverages historical crop yield data along with critical environmental parameters such as rainfall, temperature, humidity, soil moisture, and vegetation indices (NDVI). By employing advanced algorithms capable of learning both linear and non-linear patterns, the system predicts expected crop yield under varying climatic conditions.
- The proposed system serves as a decision-support tool for farmers, agricultural planners, and policymakers aiming to mitigate climate-related risks and ensure stable crop production in an increasingly uncertain environment.

Purpose

- Predict crop yield before harvest using machine learning
- Analyze key climate factors like rainfall, temperature, humidity, and soil moisture
- Detect climate risks such as drought, heat stress, and extreme weather
- Generate a climate risk score (Low/Medium/High)
- Provide early warning signals for better preparedness
- Reduce uncertainty in agricultural production
- Minimize financial losses for farmers
- Support better crop planning and selection
- Optimize use of water, fertilizers, and other resources
- Assist policymakers in agricultural planning and food security
- Improve market forecasting and supply chain planning



Literature Survey

Base paper 1:

Title: Machine Learning Frameworks for Crop Yield Forecasting

Authors: Sharma et al., Computers and Electronics in Agriculture (2023)

Draw backs:

- Does not include soil moisture data
- Does not use NDVI / vegetation indices
- Ignores soil nutrients and soil fertility factors
- No real-time data integration
- Does not assess climate risks (drought, flood, heatwave)
- Risk of overfitting (Random Forest)
- SVM not scalable for large datasets
- Limited model interpretability
- May not generalize well under changing climate conditions

Base Paper 2:

Title: Deep Learning for Agricultural Time-Series Analysis

Authors: Chen et al., IEEE Access (2024)

Draw Backs:

- Computationally heavy (high training time and resource requirements)
- Requires large historical datasets for good performance
- Not suitable for low-resource or small-scale farming systems
- Lacks interpretability (black-box model)
- No categorized climate risk assessment score
- Difficult for farmers and policymakers to understand predictions
- Sensitive to hyperparameter tuning
- Risk of overfitting with limited data
- Hard to deploy in real-time systems without strong hardware support

Existing System

- Uses traditional Machine Learning models like Random Forest, SVM, and LSTM
- Primarily relies on historical rainfall and temperature data
- Yield prediction and climate risk assessment are handled separately
- Climate risk index is not directly connected to yield prediction output
- Limited environmental parameters (often excludes soil moisture, NDVI, soil nutrients)
- No real-time weather data integration
- Limited interpretability for farmers
- Deep learning models require high computational resources
- Not designed as a unified decision-support system
- Limited scalability for multi-state or large-scale implementation

Proposed System

➤ **Advanced Machine Learning Models**

It uses advanced models such as Random Forest, LSTM, or hybrid approaches to improve prediction accuracy and capture both non-linear and time-series pattern

➤ **Multi-Feature Input Data**

The system considers multiple parameters including rainfall, temperature, soil moisture, soil nutrients (N, P, K), humidity, and NDVI to enhance reliability.

➤ **Categorized Climate Risk Score**

It generates a clear climate risk category such as Low, Medium, or High to help farmers easily understand potential threats.

➤ **Direct Link Between Risk and Yield**

The climate risk score is directly connected to the predicted crop yield, showing how environmental risk impacts production.

➤ **Real-Time Data Integration**

The system can integrate real-time weather data to provide updated and dynamic predictions.

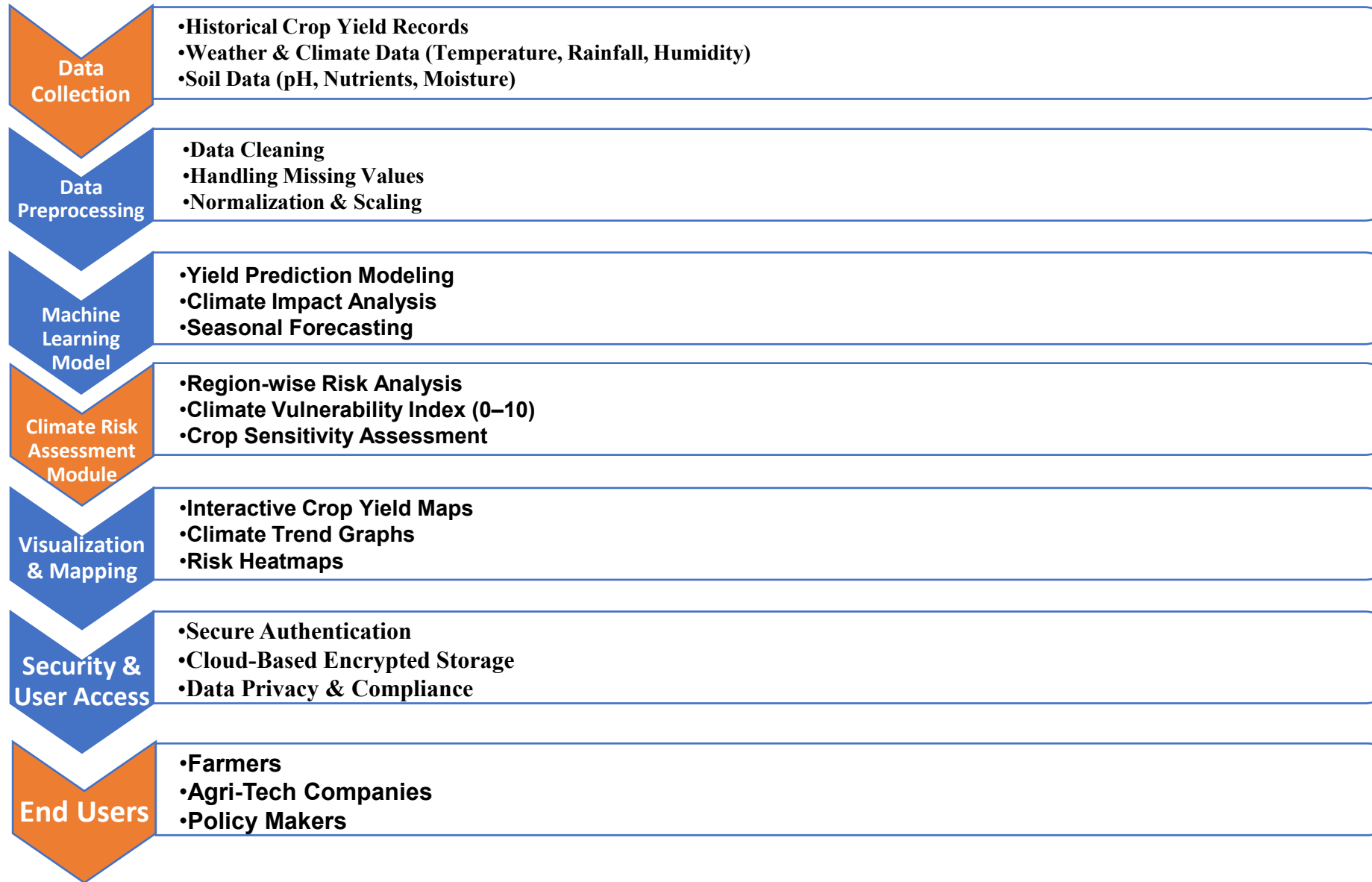
➤ **Multi-Region Support**

It supports forecasting for multiple states or regions, making it scalable for large-scale agricultural planning.

➤ **Decision-Support System**

The proposed system acts as a proactive decision-support tool, helping farmers plan irrigation, crop selection, and risk management strategies effectively.

Architectural Diagram



Flow Chart



Software Development

Development Environment

- Python (Core Programming Language)
- VS Code / Jupyter Notebook
- Git for Version Control

Libraries & Frameworks

- **Pandas, NumPy** Data Processing & Numerical Computation
- **Scikit-learn** Machine Learning Modeling
- **XGBoost / Random Forest** Crop Yield Prediction Models
- **Matplotlib / Seaborn / Plotly** Data Visualization

Data & Model Engineering

- Data Cleaning & Feature Engineering Pipeline
- Temporal & Spatial Feature Extraction
- Model Evaluation (Accuracy, Precision, Recall, F1-Score)

Frontend / Interface Development

- Interactive dashboard for input and output display.
- Real-time prediction form.
- Yield visualization charts.
- Climate risk indicator display.
- Recommendation module for farmers.

Project Implementation

➤ Data Collection

- Historical crop yield data along with climate parameters such as rainfall, temperature, humidity, soil type, and fertilizer usage were collected from agricultural and meteorological datasets. These datasets were combined to create a structured dataset suitable for analysis and modeling.

➤ Data Preprocessing

- The collected data was cleaned by handling missing values, removing duplicates, correcting inconsistencies, and managing outliers. Categorical variables such as soil type and crop type were converted into numerical form, and the dataset was standardized for machine learning processing.

➤ Feature Engineering

- Important features such as rainfall deviation, temperature anomaly, seasonal averages, and a climate risk index were generated to improve prediction accuracy. Temporal trends and region-wise climate variations were also considered to enhance the model's performance.

➤ Model Development

- A Random Forest Regressor or XGBoost algorithm was used to build the crop yield prediction model, which outputs the expected yield in tons per hectare. Additionally, a classification model was implemented to determine climate risk levels categorized as Low, Medium, or High based on climate variability indicators.

➤ Model Training and Evaluation

- The dataset was divided into training and testing sets, typically in an 80:20 ratio. Cross-validation and hyperparameter tuning were performed to improve model reliability. The regression model was evaluated using R^2 score, MAE, and RMSE, while the classification model was evaluated using accuracy and related metrics.

Reference

- Food and Agriculture Organization (FAO). (2023). *FAOSTAT Agricultural Data Portal*. Retrieved from <https://www.fao.org/faostat/>
(Used for global crop production and yield statistics.)
- India Meteorological Department (IMD). (2023). *Climate and Rainfall Data Reports*. Government of India.
(Used for rainfall and temperature datasets.)
- Ministry of Agriculture & Farmers Welfare (2023). *Agricultural Statistics at a Glance*. Government of India.
(Used for state-wise crop yield data.)
- World Bank (2022). *Climate Change Knowledge Portal*.
(Used for climate variability and risk indicators.)

THANK YOU